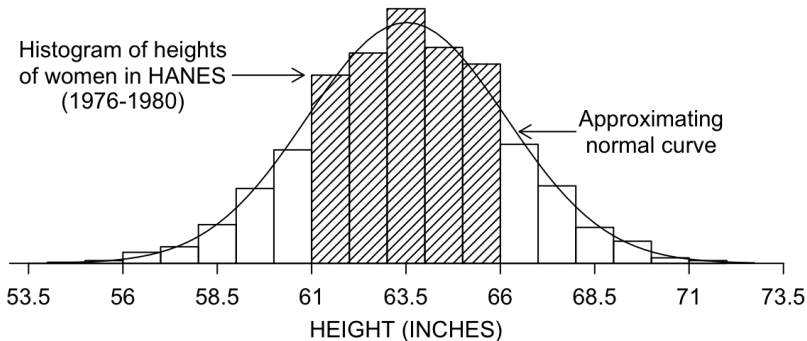


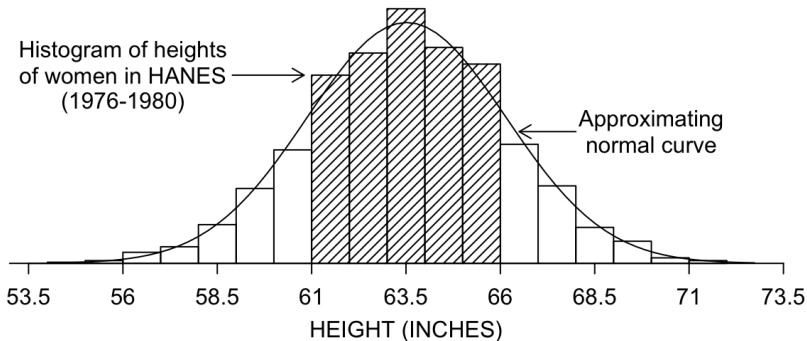
STAT 20000 Chapter 5
The Normal Approximation for Data

When a histogram is bell-shaped, the *approximate* values of percentile can be found in a *simple* way
— approximating the histogram by a **normal curve**.



Why Are Normal Curves of Interest?

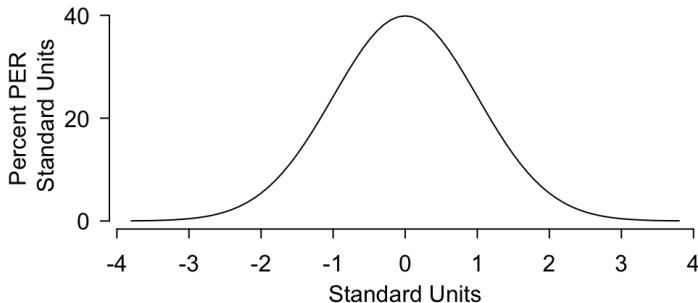
1. Many variables (e.g., height, blood pressure, . . . , but not years of education, . . .) have histograms which follow (match up well with) a normal curve.
2. For such variables, areas under the histogram — that is, population percentages — can be approximated by the corresponding areas under the normal curve:



Why Are Normal Curves of Interest? (Cont'd)

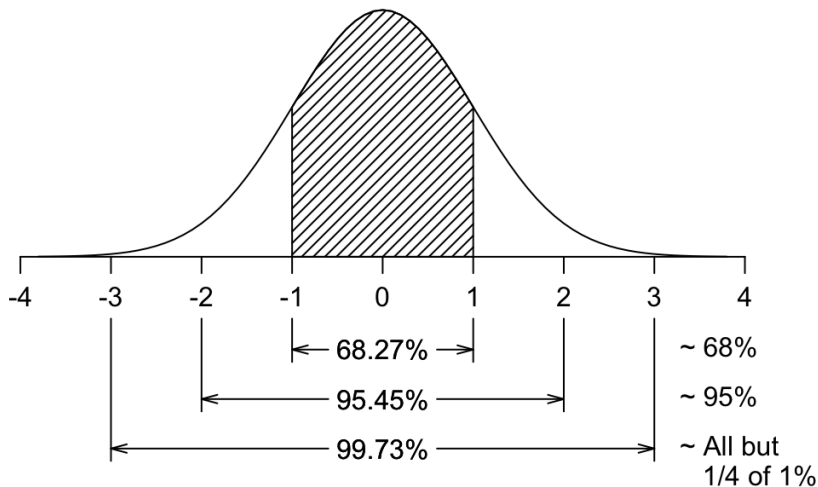
3. Areas under the normal curve can be computed easily knowing only the *average* and the *SD*.
4. Normal curves are well known and well understood.
 - ▶ A convenient means of communication.
5. The probability histogram of the averages of large samples tends to follow the normal curve (Central Limit Theorem, Chapter 18).
 - ▶ The cornerstone of statistical inference!

The Standard Normal Curve



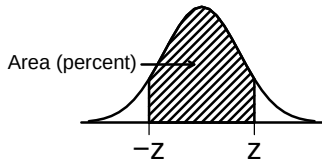
- ▶ The equation of the curve: $y = \frac{100\%}{\sqrt{2\pi}} e^{-x^2/2}$
- ▶ Two very important properties are:
 1. The total area under the curve is **100%**,
— just like a histogram.
 2. The curve is **symmetric** about 0.

A Brief Table of Areas under the Standard Normal Curve



Normal Table

FPP, page A-155



We will **NOT** use this FPP normal table.

We will use the more commonly used normal table on the next page instead!

z	Area	z	Area	z	Area
0.00	0.	1.50	86.64	3.00	99.730
0.05	3.99	1.55	87.89	3.05	99.771
0.10	7.97	1.60	89.04	3.10	99.806
0.15	11.92	1.65	90.11	3.15	99.837
0.20	15.85	1.70	91.09	3.20	99.863
0.25	19.74	1.75	91.99	3.25	99.885
0.30	23.58	1.80	92.81	3.30	99.903
0.35	27.37	1.85	93.57	3.35	99.919
0.40	31.08	1.90	94.26	3.40	99.933
0.45	34.73	1.95	94.88	3.45	99.944
0.50	38.29	2.00	95.45	3.50	99.953
0.55	41.77	2.05	95.96	3.55	99.961
0.60	45.15	2.10	96.43	3.60	99.968
0.65	48.43	2.15	96.84	3.65	99.974
0.70	51.61	2.20	97.22	3.70	99.978
0.75	54.67	2.25	97.56	3.75	99.982
0.80	57.63	2.30	97.86	3.80	99.986
0.85	60.47	2.35	98.12	3.85	99.988
0.90	63.19	2.40	98.36	3.90	99.990
0.95	65.79	2.45	98.57	3.95	99.992
1.00	68.27	2.50	98.76	4.00	99.9937
1.05	70.63	2.55	98.92	4.05	99.9949
1.10	72.87	2.60	99.07	4.10	99.9959
1.15	74.99	2.65	99.20	4.15	99.9967
1.20	76.99	2.70	99.31	4.20	99.9973
1.25	78.87	2.75	99.40	4.25	99.9979
1.30	80.64	2.80	99.49	4.30	99.9983
1.35	82.30	2.85	99.56	4.35	99.9986
1.40	83.85	2.90	99.63	4.40	99.9989
1.45	85.29	2.95	99.68	4.45	99.9991

Normal probability table



Z	Second decimal place of Z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8728	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

*For $Z \geq 3.50$, the probability is greater than or equal to 0.9998.

Normal probability table



Z	Second decimal place of Z									
	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.00
.0002	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003
.0003	.0004	.0004	.0004	.0004	.0004	.0004	.0004	.0005	.0005	.0005
.0005	.0005	.0005	.0005	.0006	.0006	.0006	.0006	.0006	.0007	.0007
.0007	.0007	.0008	.0008	.0008	.0008	.0008	.0009	.0009	.0009	.0010
.0010	.0010	.0011	.0011	.0011	.0011	.0012	.0012	.0013	.0013	.0013
.0014	.0014	.0015	.0015	.0016	.0016	.0016	.0017	.0018	.0018	.0019
.0019	.0020	.0021	.0021	.0022	.0022	.0023	.0023	.0024	.0025	.0026
.0026	.0027	.0028	.0029	.0030	.0031	.0031	.0032	.0033	.0034	.0035
.0036	.0037	.0038	.0039	.0040	.0041	.0041	.0043	.0044	.0045	.0047
.0048	.0049	.0051	.0052	.0054	.0055	.0057	.0059	.0060	.0062	.0062
.0064	.0066	.0068	.0069	.0071	.0073	.0075	.0078	.0080	.0082	.0084
.0084	.0087	.0089	.0091	.0094	.0096	.0099	.0102	.0104	.0107	.0107
.0110	.0113	.0116	.0119	.0122	.0125	.0129	.0132	.0136	.0139	.0141
.0143	.0146	.0150	.0154	.0158	.0162	.0166	.0170	.0174	.0179	.0181
.0183	.0188	.0192	.0197	.0202	.0207	.0212	.0217	.0222	.0228	.0230
.0233	.0239	.0244	.0250	.0256	.0262	.0268	.0274	.0281	.0287	.0290
.0294	.0301	.0307	.0314	.0322	.0329	.0336	.0344	.0351	.0359	.0361
.0367	.0375	.0384	.0392	.0401	.0409	.0418	.0427	.0436	.0446	.0449
.0455	.0465	.0475	.0485	.0495	.0505	.0516	.0526	.0537	.0548	.0551
.0559	.0571	.0582	.0594	.0606	.0618	.0630	.0643	.0655	.0668	.0671
.0681	.0694	.0708	.0721	.0735	.0749	.0764	.0778	.0793	.0808	.0812
.0823	.0838	.0853	.0869	.0885	.0901	.0918	.0934	.0951	.0968	.0972
.0985	.1003	.1020	.1038	.1056	.1075	.1093	.1112	.1131	.1151	.1155
.1170	.1190	.1210	.1230	.1251	.1271	.1292	.1314	.1335	.1357	.1361
.1379	.1401	.1423	.1446	.1469	.1492	.1515	.1539	.1562	.1587	.1591
.1611	.1635	.1660	.1685	.1711	.1736	.1762	.1788	.1814	.1841	.1845
.1867	.1894	.1922	.1949	.1977	.2005	.2033	.2061	.2090	.2119	.2123
.2148	.2177	.2206	.2236	.2266	.2296	.2327	.2358	.2389	.2420	.2424
.2451	.2483	.2514	.2546	.2578	.2611	.2643	.2676	.2709	.2743	.2747
.2776	.2810	.2843	.2877	.2912	.2946	.2981	.3015	.3050	.3085	.3089
.3121	.3156	.3192	.3228	.3264	.3300	.3336	.3372	.3409	.3446	.3450
.3483	.3520	.3557	.3594	.3632	.3669	.3707	.3745	.3783	.3821	.3825
.3859	.3897	.3936	.3974	.4013	.4052	.4090	.4129	.4168	.4207	.4211
.4247	.4286	.4325	.4364	.4404	.4443	.4483	.4522	.4562	.4602	.4606
.4641	.4681	.4721	.4761	.4801	.4840	.4880	.4920	.4960	.5000	.5000

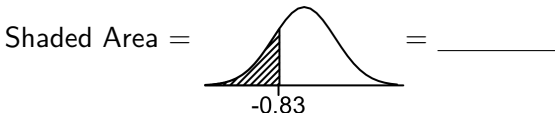
*For $Z \leq -3.50$, the probability is less than or equal to 0.0002.

The *normal probability table* on the previous slide gives the *area under the standard normal curve to the left of z* .



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

E.g., for $z = -0.83$, look at the row -0.8 and the column 0.03 .

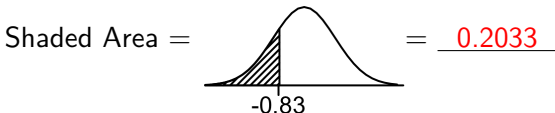


The *normal probability table* on the previous slide gives the *area under the standard normal curve to the left of z* .



<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

E.g., for $z = -0.83$, look at the row -0.8 and the column 0.03 .

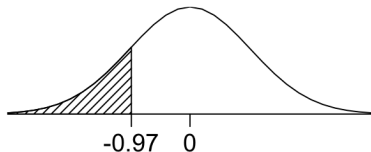


In-Class Exercise 1a

Find the area under the standard normal curve to the left of -0.97 .

In-Class Exercise 1a

Find the area under the standard normal curve to the left of -0.97 .



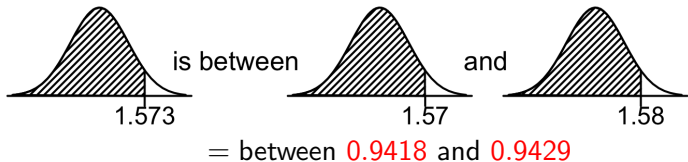
	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.00
-1	0.1379	0.1401	0.1423	0.1446	0.1469	0.1492	0.1515	0.1539	0.1562	0.1587
-0.9	0.1611	0.1635	0.1660	0.1685	0.1711	0.1736	0.1762	0.1788	0.1814	0.1841
-0.8	0.1867	0.1894	0.1922	0.1949	0.1977	0.2005	0.2033	0.2061	0.2090	0.2119

Ans: 0.1660

Question: Find the **area** under the standard normal curve to the **left** of 1.573.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

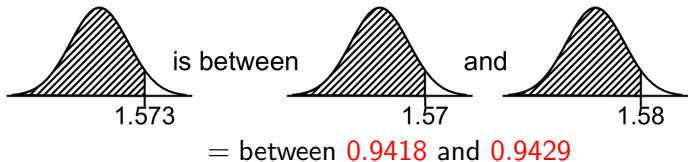


All values between 0.9418 and 0.9429 are acceptable in HWs and exams.

Question: Find the **area** under the standard normal curve to the **left** of 1.573.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮



All values between 0.9418 and 0.9429 are acceptable in HWs and exams.

Finding *Upper* Tail Area Under Standard Normal Curve

What is the area under the standard normal curve to the right of -0.83 ?

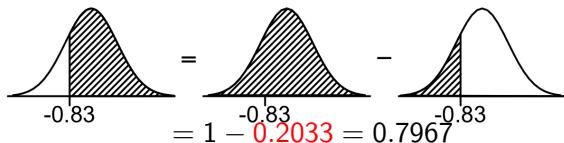
upper-tail



Finding *Upper* Tail Area Under Standard Normal Curve

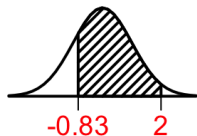
What is the area under the standard normal curve to the right of -0.83 ?

upper-tail

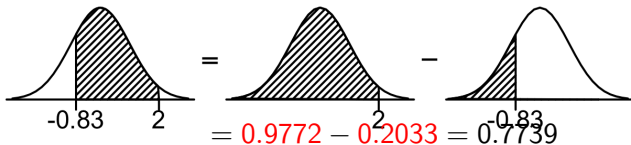


<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148

Question: Find the **area** under the standard normal curve **between** -0.83 and 2 .



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857

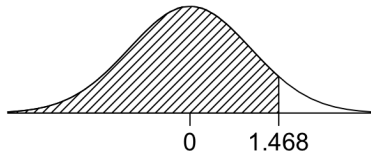


In-Class Exercise 1b

Find the area under the standard normal curve to the left of 1.468.

In-Class Exercise 1b

Find the area under the standard normal curve to the left of 1.468.



	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441

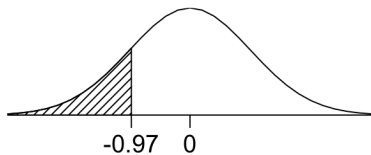
Ans: About 0.93 (Any value between 0.9292 and 0.9306 is acceptable)

In-Class Exercise 1c

Find the area under the standard normal curve to the right of -0.97

In-Class Exercise 1c

Find the area under the standard normal curve to the right of -0.97



	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.00
-1	0.1379	0.1401	0.1423	0.1446	0.1469	0.1492	0.1515	0.1539	0.1562	0.1587
-0.9	0.1611	0.1635	0.1660	0.1685	0.1711	0.1736	0.1762	0.1788	0.1814	0.1841
-0.8	0.1867	0.1894	0.1922	0.1949	0.1977	0.2005	0.2033	0.2061	0.2090	0.2119

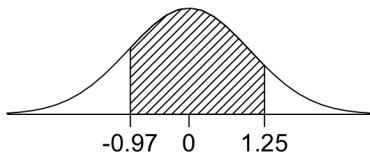
$$\text{Ans} = 1 - 0.1660 = 0.8340.$$

In-Class Exercise 1d

Find the area under the standard normal curve between -0.97 and 1.25 .

In-Class Exercise 1d

Find the area under the standard normal curve between -0.97 and 1.25 .



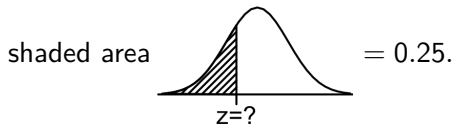
	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.00
-1	0.1379	0.1401	0.1423	0.1446	0.1469	0.1492	0.1515	0.1539	0.1562	0.1587
-0.9	0.1611	0.1635	0.1660	0.1685	0.1711	0.1736	0.1762	0.1788	0.1814	0.1841
-0.8	0.1867	0.1894	0.1922	0.1949	0.1977	0.2005	0.2033	0.2061	0.2090	0.2119

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

$$\text{Ans} = 0.8944 - 0.1660 = 0.7284.$$

Example: 25th Percentile of the Standard Normal

The 25th percentile of the standard normal is the z that

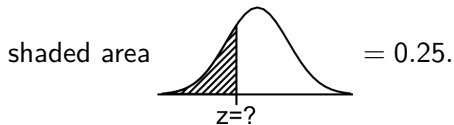


We need to search in the *body* of the normal table for the values closest to 0.25, which are 0.2514 and 0.2483, and find the z corresponding to those values, which are -0.67 and -0.68 .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Example: 25th Percentile of the Standard Normal

The 25th percentile of the standard normal is the z that

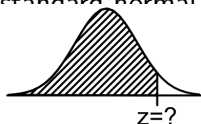


We need to search in the *body* of the normal table for the values closest to 0.25, which are 0.2514 and 0.2483, and find the z corresponding to those values, which are -0.67 and -0.68 .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Example: 95th Percentile of the Standard Normal

The 95th percentile of the standard normal is the z such that shaded area = 0.95.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633

The values closest to 0.95 in the body of the normal table are 0.9495 and 0.9505, corresponding to z -values 1.64 and 1.65.

The 95th percentile is thus between 1.64 and 1.65.

In-Class Exercise 2

Find the 30th percentile of the standard normal curve.

In-Class Exercise 2

Find the 30th percentile of the standard normal curve.

	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.00
-0.6	0.2451	0.2483	0.2514	0.2546	0.2578	0.2611	0.2643	0.2676	0.2709	0.2743
-0.5	0.2776	0.2810	0.2843	0.2877	0.2912	0.2946	0.2981	0.3015	0.3050	0.3085
-0.4	0.3121	0.3156	0.3192	0.3228	0.3264	0.3300	0.3336	0.3372	0.3409	0.3446

The 30th percentile is thus between -0.52 and -0.53 .

In-Class Exercise 3

Find the 82nd percentile of the standard normal curve.

In-Class Exercise 3

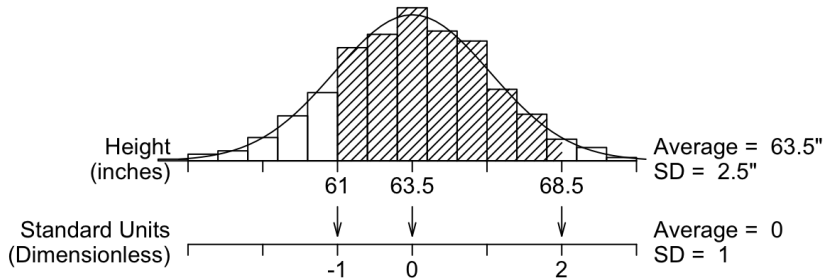
Find the 82nd percentile of the standard normal curve.

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621

The 82nd percentile is thus between 0.91 and 0.92.

The Normal Approximation, An Example

Consider the heights of women in HANES:



What percentage of women are between 61 and 68.5 inches tall?

Standard Units

Standard units say how many SDs a value is above (+ sign) or below (− sign) average.

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

1. What is 61" in standard units?

- ▶ 61" is _____ inches _____ average.
- ▶ That's _____ SD _____ average.
- ▶ So 61" is _____ in standard units.

2. What is 68.5" in standard units?

- ▶ 68.5" is _____ inches _____ average.
- ▶ That's _____ SD _____ average.
- ▶ So 68.5" is _____ in standard units.

Standard Units

Standard units say how many SDs a value is above (+ sign) or below (− sign) average.

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

1. What is 61" in standard units?

- ▶ 61" is 2.5 inches below average.
- ▶ That's _____ SD _____ average.
- ▶ So 61" is _____ in standard units.

2. What is 68.5" in standard units?

- ▶ 68.5" is _____ inches _____ average.
- ▶ That's _____ SD _____ average.
- ▶ So 68.5" is _____ in standard units.

Standard Units

Standard units say how many SDs a value is above (+ sign) or below (− sign) average.

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

1. What is 61" in standard units?

- ▶ 61" is 2.5 inches below average.
- ▶ That's 1 SD below average.
- ▶ So 61" is _____ in standard units.

2. What is 68.5" in standard units?

- ▶ 68.5" is _____ inches _____ average.
- ▶ That's _____ SD _____ average.
- ▶ So 68.5" is _____ in standard units.

Standard Units

Standard units say how many SDs a value is above (+ sign) or below (− sign) average.

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

1. What is 61" in standard units?

- ▶ 61" is 2.5 inches below average.
- ▶ That's 1 SD below average.
- ▶ So 61" is −1 in standard units.

2. What is 68.5" in standard units?

- ▶ 68.5" is _____ inches _____ average.
- ▶ That's _____ SD _____ average.
- ▶ So 68.5" is _____ in standard units.

Standard Units

Standard units say how many SDs a value is above (+ sign) or below (− sign) average.

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

1. What is 61" in standard units?

- ▶ 61" is 2.5 inches below average.
- ▶ That's 1 SD below average.
- ▶ So 61" is −1 in standard units.

2. What is 68.5" in standard units?

- ▶ 68.5" is 5 inches above average.
- ▶ That's SD average.
- ▶ So 68.5" is in standard units.

Standard Units

Standard units say how many SDs a value is above (+ sign) or below (− sign) average.

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

1. What is 61" in standard units?

- ▶ 61" is 2.5 inches below average.
- ▶ That's 1 SD below average.
- ▶ So 61" is −1 in standard units.

2. What is 68.5" in standard units?

- ▶ 68.5" is 5 inches above average.
- ▶ That's 2 SD above average.
- ▶ So 68.5" is in standard units.

Standard Units

Standard units say how many SDs a value is above (+ sign) or below (− sign) average.

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

1. What is 61" in standard units?

- ▶ 61" is 2.5 inches below average.
- ▶ That's 1 SD below average.
- ▶ So 61" is −1 in standard units.

2. What is 68.5" in standard units?

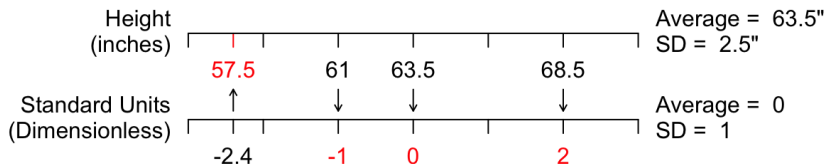
- ▶ 68.5" is 5 inches above average.
- ▶ That's 2 SD above average.
- ▶ So 68.5" is +2 in standard units.

Standard Units (Cont'd)

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

3. What height is -2.4 in standard units?

- ▶ The height is _____ SDs _____ average.
- ▶ That's _____ inches _____ average.
- ▶ The height is _____.

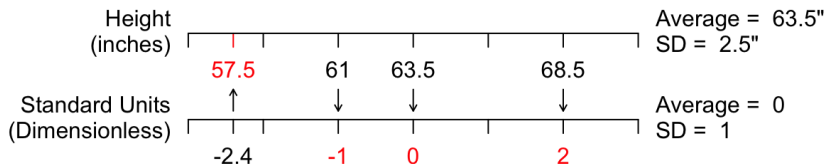


Standard Units (Cont'd)

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

3. What height is -2.4 in standard units?

- ▶ The height is 2.4 SDs _____ average.
- ▶ That's _____ inches _____ average.
- ▶ The height is _____.

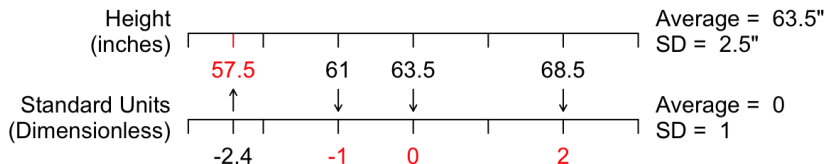


Standard Units (Cont'd)

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

3. What height is -2.4 in standard units?

- ▶ The height is 2.4 SDs below average.
- ▶ That's _____ inches _____ average.
- ▶ The height is _____.

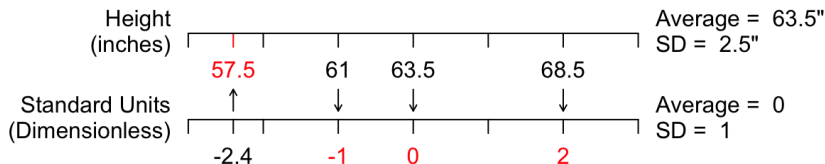


Standard Units (Cont'd)

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

3. What height is -2.4 in standard units?

- ▶ The height is 2.4 SDs below average.
- ▶ That's $2.4 \times 2.5 = 6$ inches below average.
- ▶ The height is _____.

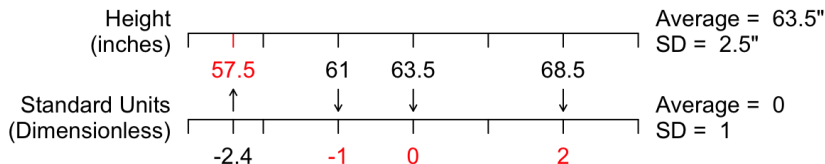


Standard Units (Cont'd)

Example: the women in the HANES study had heights averaging to 63.5", with an SD of 2.5".

3. What height is -2.4 in standard units?

- ▶ The height is 2.4 SDs below average.
- ▶ That's $2.4 \times 2.5 = 6$ inches below average.
- ▶ The height is $63.5 - 6 = 57.5$.



Original Scale $\Leftrightarrow$ Standard-Unit Scale

How to convert a value to standard units?

$$\text{standard units} = \frac{\text{value} - \text{average}}{\text{SD}}.$$

E.g., to express 68.5" in standard units, you compute

$$\frac{68.5'' - 63.5''}{2.5''} = \frac{5.0''}{2.5''} = 2$$

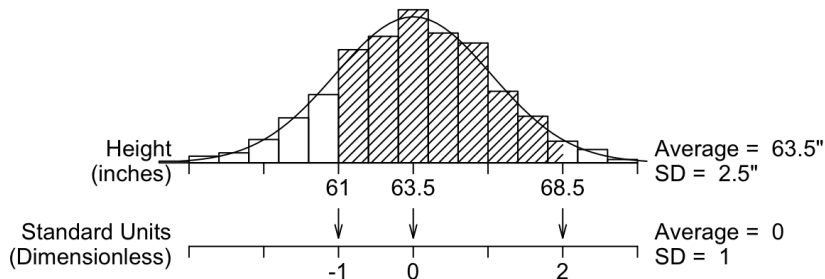
How to convert to the original scale back from standard units?

$$\text{value} = \text{average} + (\text{standard units} \times \text{SD}).$$

E.g., to find the height corresponding to -2.4 in standard units, you compute

$$63.5'' + (-2.4 \times 2.5'') = 63.5'' - 6'' = 57.5''$$

Back to the Normal Approximation Example



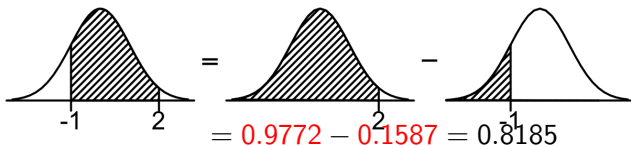
The percentage of women between 61" and 68.5" tall is

- ▶ exactly equal to the area under the histogram from 61" to 68.5", and
- ▶ approximately equal to the area under the normal curve between -1 and 2. See next page.

It remains to find the **area** under the standard normal curve **between** -1 and 2 .



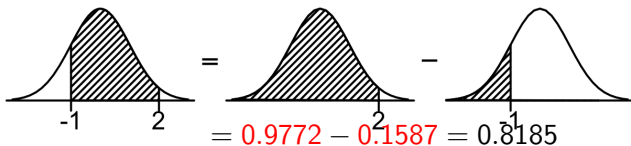
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857



It remains to find the **area** under the standard normal curve **between** -1 and 2 .



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857



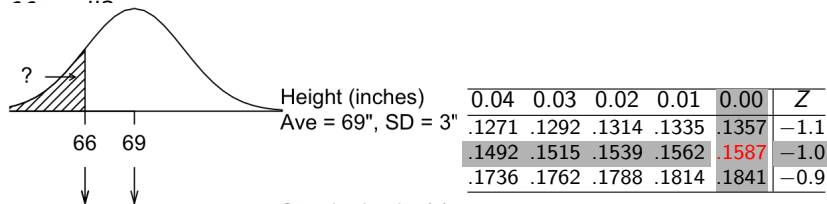
Answer: 81.85%

The Normal Approximation (Cont'd)

- ▶ If a list of numbers follows the normal curve, the percentage of entries falling in a given interval can be estimated by first converting the interval to standard units and then finding the corresponding area under the standard normal curve. This procedure is called the **normal approximation**.
- ▶ If a histogram follows the normal curve,
 - ▶ about 68% of the area lies within one SD of the average, and
 - ▶ about 95% within two SDs of the average.
- ▶ **Warning:** The normal approximation is only valid if the histogram is approximately normal. Use your judgement.

Using the Normal Approximation

A group of men have heights that follow a normal curve with average 69" and SD 3. About what percentage of them are below

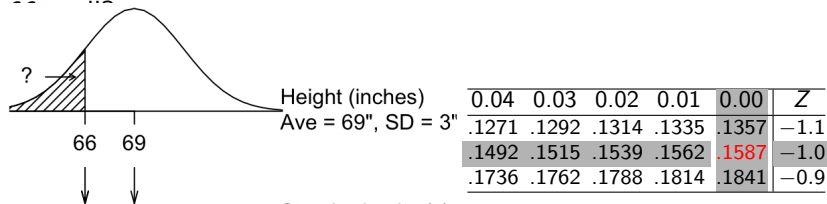


1. Sketch a normal curve, and shade area of interest
2. Convert 66" to standard unit: $z = \frac{66 - 69}{3} = -1$
3. From Normal Table, the area to the left of -1 is **0.1587**.

Answer: 15.87% or about 16%.

Using the Normal Approximation

A group of men have heights that follow a normal curve with average 69" and SD 3. About what percentage of them are below



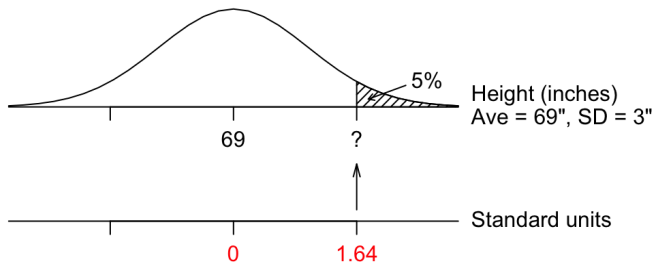
1. Sketch a normal curve, and shade area of interest
2. Convert 66" to standard unit: $z = \frac{66 - 69}{3} = -1$
3. From Normal Table, the area to the left of -1 is **0.1587**.

Answer: 15.87% or about 16%.

Method: original units $\rightarrow$ standard units $\rightarrow$ normal curve.

Same population (heights averaging to 69" with an SD of 3").

How tall a man must be to be taller than 95% of the population?



Thus

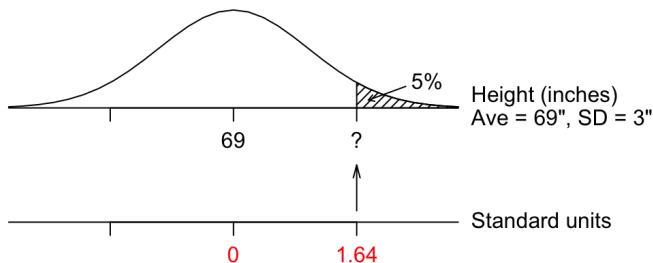
$$\begin{aligned} ? = \text{height} &= \text{_____ SDs above average} \\ &= 1.64 \times 3'' + 69'' \\ &= 5'' + 69'' = 74'' = 6'2'' \end{aligned}$$

The method:

standard normal curve $\rightarrow$ standard units $\rightarrow$ original units.

Same population (heights averaging to 69" with an SD of 3").

How tall a man must be to be taller than 95% of the population?



Thus

$$\begin{aligned} ? = \text{height} &= \underline{1.64} \text{ SDs above average} \\ &= 1.64 \times 3'' + 69'' \\ &= 5'' + 69'' = 74'' = 6'2'' \end{aligned}$$

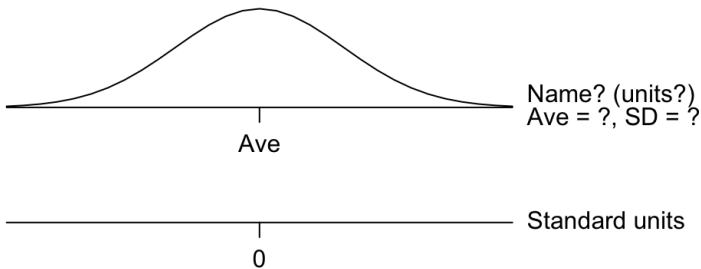
The method:

standard normal curve $\rightarrow$ standard units $\rightarrow$ original units.

The general procedure for working these kinds of problems:

Draw the picture!

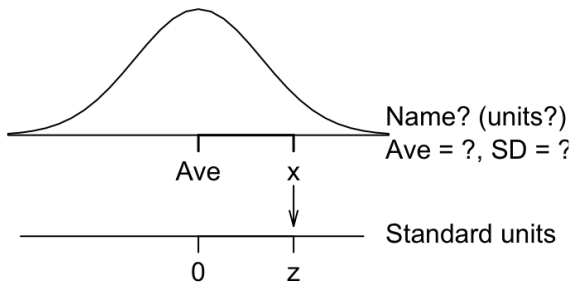
- ▶ Sketch the normal curve
- ▶ Put in the axis for the original units
- ▶ Put in the axis for the standard units
- ▶ Shade the area of interest
- ▶ Proceed



Follow this procedure on the HW, quizzes, and exams!

General Procedure for Finding Normal Areas

1. Sketch the normal curve
2. Put in the axis for the original units
3. Put in the axis for the standard units
4. Shade the area of interest
5. Convert end point (values) to standard units (z)
6. Find lower-tail area(s) for standard units from the normal table
7. Calculate the area of interest



General Procedure for Finding Normal Percentiles

1. Sketch the normal curve
2. Put in the axis for the original variable X
3. Put in the axis for the standard units
4. Shade the area of interest
5. Find the standard unit z for the lower-tail area from the normal table
6. Find original value X (= percentile) from standard unit z

